Experiment No. 4

1. Aim: Design and implement a CNN model for image classification.

2. Objectives: From this experiment, the student will be able to

Learn different Deep Neural Networks

3. Outcomes: Learner will be able to understand the working of CNN on image data.

4. Software Required: Python

5. Theory:

Image classification is a fundamental problem in computer vision, where an algorithm assigns a label to an input image based on its content. Traditional machine learning methods rely on handcrafted features, whereas Convolutional Neural Networks (CNNs) automatically learn feature representations from raw image data. CNNs are widely used for tasks such as object detection, facial recognition, and medical imaging. The design and implementation of a Convolutional Neural Network (CNN) for image classification. The model is trained on the Fashion MNIST dataset, which consists of grayscale images representing different clothing categories. The CNN architecture includes convolutional layers, pooling layers, and fully connected layers to extract hierarchical features and classify images accurately. The model is implemented using TensorFlow and Keras.

- CNNs automatically learn important features from image data.
  a. ReLU activation prevents vanishing gradients.
  b. Softmax activation is used for multi-class classification.
  c. Adam optimizer provides efficient training convergence.
  d. Model generalizes well with minimal overfitting.

6. Design:

a. **CNN model** consists of multiple layers designed to extract patterns from images. The architecture used in this study includes:
b. **Convolutional Layers:** Learn spatial hierarchies of features by applying filters to the input image.
c. **Pooling Layers:** Reduce spatial dimensions, making computation efficient and reducing overfitting.
d. **Fully Connected Layers**: Combine extracted features for classification.

e. **Activation Functions:** Introduce non-linearity (ReLU in hidden layers, Softmax in the output layer).

f. **Loss Function:** Categorical cross-entropy to measure classification performance.

7. Procedure Steps:

a) Import Libraries

b) Define CNN Model

c) Compile and Train the Model

d) Evaluate Model

e) Results & Discussion

8. Conclusion/ Results:

This program is successfully implemented with a CNN model for image classification using the Fashion MNIST dataset**.** The model achieved high accuracy by utilizing convolutional layers to extract features, followed by fully connected layers for classification. Future improvements could include data augmentation and transfer learning to further enhance performance.

9. Viva Question:

a) How to use CNN for image classification?
b) What is CNN classifier for image classification??
c) How do I denoise an image in CNN?
d) Can CNN be used for image segmentation?

10. References:

1. Francois Chollet, “Deep learning with Python” – Manning Publications.
2. Ian Goodfellow, Yoshua Bengio, Aaron Courville. ―Deep Learning‖, MIT Press Ltd, 2016.
3. https://www.kaggle.com/code/arbazkhan971/image-classification-using-cnn-94-accuracy.

4. https://www.analyticsvidhya.com/blog/2020/02/learn-image-classification-cnn-convolutional-neural-networks-3-datasets/